MID-TERM EXAMINATION

(CBCS)

B. Tech (CSAI/ECEAI), 2nd Semester

(June, 2022)

Subject Code: BAS 108	Subject: Probability and Statistics
Time : 1 ½ Hours	Maximum Marks : 30
Note:Q. 1 is compulsory. Attempt any two questions from the rest.	

Q1		(2.5*4)
	(a) Let X and Y be independent random variables with $[X] = 3, E[X^2] = 25, E[Y] = 10, E[Y^2] = 164$. Find $E[2X - 3Y + 7]$and $Var[2X - 3Y + 7]$.	
	(b) In a certain college, 25% of boys and 20% of girls are studying mathematics. The girls constitute 60% of the student body. If a student is selected at random and is found studying mathematics, find the probability that the student is a girl?	
	(c) The joint distribution function of X and Y be $$f(x,y) = \begin{cases} \frac{1}{n^2}, & x = 1,2,3,\dots\dots,n \text{ and } y = 1,2,3,\dots\dots,n \end{cases}$$ Find the marginal density for Y.	
	(d) Prove that $Cov(aX + b, cY + d) = acCov(X,Y)$ for constants a, b, c, d.	
Q2		**(5,5)**
	(a) The joint density for (X,Y) is given as $$f(x,y) = \begin{cases} 15x^2y & 0 < x < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$ Find $E[XY]$.	
	(b) Let X be a discrete random variable that assumes the values -2, -1, 0, 1 and 2.It is given that: $P(X = -2) = P(X = -1) = P(X = 1) = P(X = 2)$ and $P(X < 0) = P(X = 0) = P(X > 0) = \propto$. Determine the probability mass function and the cumulative distribution function of X.	
Q3		**(5,5)**
	(a) A continuous random variable X has density function $$f(x,y) = \begin{cases} 2e^{-2x} & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$ Find the expected value of X.	
	(b) The joint density for (X,Y) is given as $$f(x,y) = \begin{cases} 60x^2y & 0 \le x, y \le 1, \text{and } x + y \le 1 \\ 0, & \text{elsewhere} \end{cases}$$ Find the conditional probability density function of X given Y (i.e. $f(x\|y)$).	
Q4		**(5,5)**
	(a) Let X denote the length in minutes of a long distance telephone conversation. The density for X is given by $$f(x,y) = \begin{cases} \frac{1}{10}e^{-x/10}, & x > 0 \end{cases}$$ Find the moment generating function of X.	
	(b) Let the joint pdf of X and Y be $$f(x,y) = \begin{cases} K(x + y); & x > 0, y > 0, 3x + y < 3 \\ 0 \quad ; & \text{otherwise} \end{cases}$$ Find the value of the constant K.	